Aim:- To prepare the connection of Sodium vapour lamp and Halogen lamp with measure voltage and current.

Apparatus Required:-

S.No	Name	Range	QTY.
1.	Hpsv lamp	220-250V,70W	1
2.	Hpsv lamp holder	5A	1
3.	Copper ballast	220-240V,1A, P.F.=0.34	1
4.	Igniter	20000V	1
5.	Capacitor	250V,10 MICRO FARAD	1
6.	DPMCB	6A, 250V	1
7.	Leads	RED,BLACK,YELLOW	1

Theory:-

The practically lamp of this development of this type of lamp, however was delayed since ordinary glass cannot with stand the chemical action of hot sodium.

The sodium vapour lamp consists of bulbs containing a small amount of metallic sodium,to develop enough heat to vaporize the sodium.Since long discharge paths are necessary therefore, the discharge envelope is usually bent into u-shape.

The lamp operates at temp. 300 degree temp in order to conserve the heat generate and assure the lamp operating at normal air temp.

The discharge envelope is closed in a special vacuum envelop designed for this purpose. The lamp must be operated horizontally or nearly so, to keep the sodium well spread out along the tube, although some small lamp, may be operated vertically lamp up.

The sodium vapour lamp is only suitable for alternate current and therefore requires chocke control. This requirement is met by operating the lamp for a stray field step-up-tapped auto transformed auto transformed with an open circuit secondary voltage of 470-480 volts. The uncorrected power factor is very low.

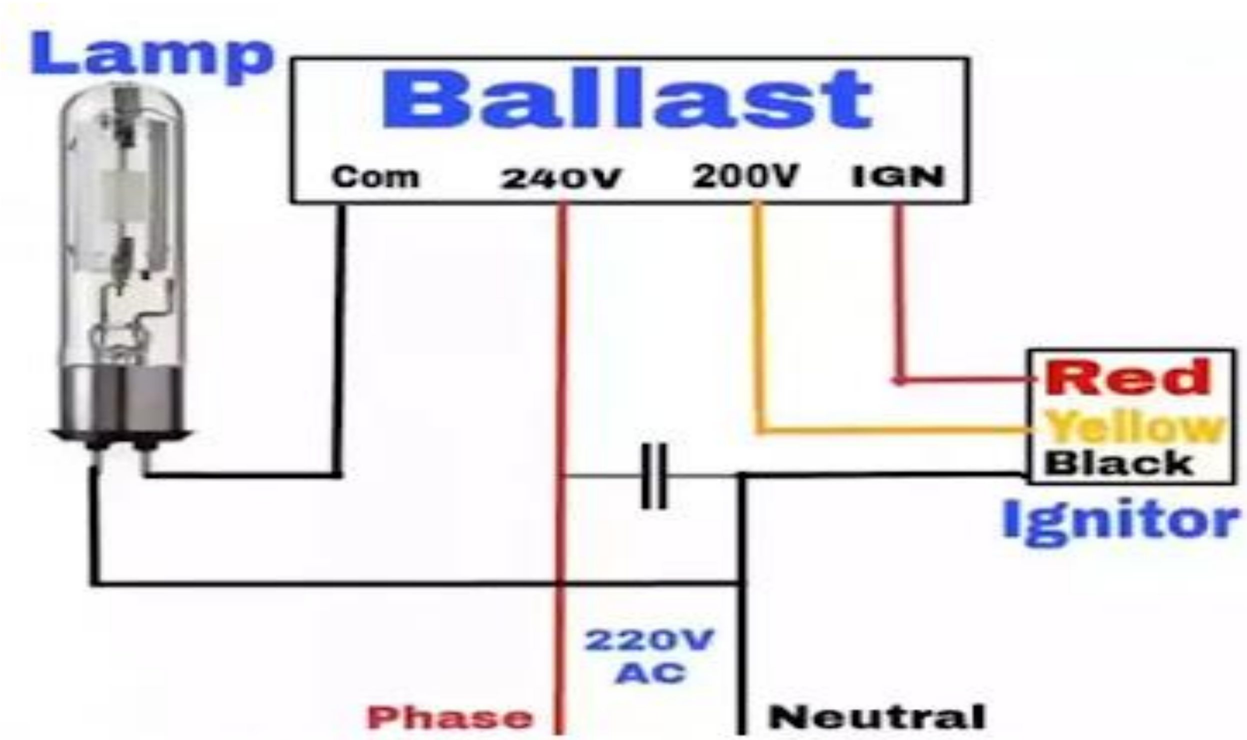
When is not in operation , the sodium is usually in the form of solid deposited on the side walls of tube, therefore at first when it is connected across the supply mains, the discharge takes places in the neon gas. The metallic sodium vaporizes and then ionizes.

The efficiency of a sodium vapour lamp under practically condition is about 40-50 lumens/watt. The major application of this type of lamp is for high way and generator outdoor lighting where colour discrimination is not required, as required, such as street lighting parks , rail yards,lamps are manufactured in 45,60,85and is not affected by voltage variation at the end of this period the light output will be reduced by 15% due to ageing.

Advantage:-

1. No dimensions of lamp miniature size.
2. Long life-20000 hours.
3. Better colour rendition.

Circuit Diagram:-



Sodium vapour lamp connection

Halogen Lamp:-

The halogen lamp is the latest member in the family of incandescent lamp. It numerous advantages over the ordinary incandescent lamp. As already stated the life and efficiency of an incandescent lamp fall of with use-partly due to slow evaporation of the filament and partly due to back deposit formed on the inside of the inside of the bulb.

The addition of small amount of halogen vapours to the filling gas restores port of the evaporated tungsten vapour back to filament by means of chemical reaction i.e. there is a sort of regeneration cycle.

Advantage:-

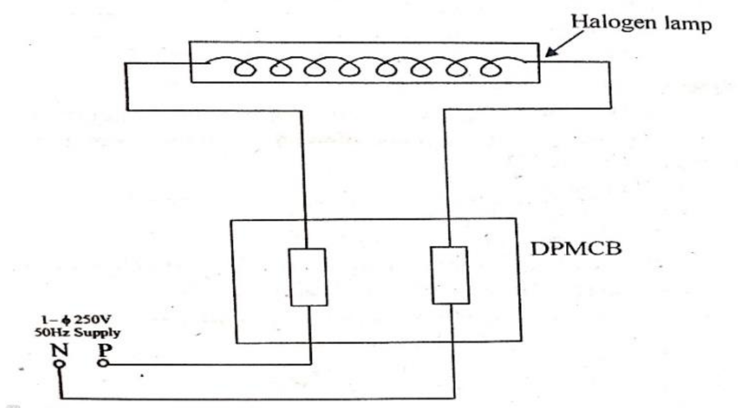
1. No dimensions of lamp miniature size.
2. Long life-20000 hours.
3. Better colour rendition.

Application:-

Halogen lamp (up to 5 KW) –

Playing Field, Large Garden,Car Parking, T.V. , Studio Etc.

Circuit Diagram:-



Precaution:-

1. Connection should be proper. Loose connection is not allowed.
2. Do not touch any live wire or part.
3. Before switch on the supply, the connection should be checked by lab assistant.
4. Do not damage the terminals of part or wire etc.

Procedure:-

1. Make the connection as per as diagram.
2. Connections check and give the supply.
3. When lamp does not glow then filament open.
4. When lamp bright glow then filament is short.
5. When dim glow then filament is in proper working condition.

Observation Table:-

S.No	Switch	Condition	Voltage(v)	Current(I)
1.	S1	on		
2.	S1	off		

Results:- we have successfully prepare the connection and study of Sodium vapour lamp and Halogen lamp with measure the Voltage and Current.

Viva- Voice:-

Que.1.) What is the function of ignitor in sodium vapour lamps?

Que.2) What is life and efficiency of SV lamps?

Que3.)How much time taken by sodium vapour lamp to full glow?

Que -4) What is the composition of inert gases used in filament lamp?

Que.-5) What is life of approx. incandesend lamp?

Viva- Voice:-

Ques 1.)Why halogen lamp popular than other lamps?

Ques 2.) What is diff. b/w resistivity and specific resistance?

Ques 3.)What do you mean by resistor?

Que -4) What is the composition of inert gases used in filament lamp?

Que.-5) What is life of approx. incandesend lamp?